

Cheng Yuan (Ross) King

MSc AI | Data & ML Engineering

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

CAREER ASPIRATION

I am an MSc Artificial Intelligence student who builds data and ML systems whose decisions you can audit. My experience runs across data engineering, analytics engineering and applied ML: modelling warehouses in dbt, moving data at scale with Spark, and putting models behind tested, deployed services. I am skilled in SQL, Python, dbt and the cloud data stack. I am available for full-time contract work from October 2026, and for permanent roles from December once I switch to the UK Graduate visa — two years, no sponsorship needed.

SKILLS

Programming	Python, SQL, Bash, TypeScript
Data Engineering	PySpark, dbt, Dagster, DuckDB/Parquet, PostgreSQL, BigQuery, Snowflake, GitHub Actions
ML & Modelling	scikit-learn, LightGBM, calibration, forecasting, anomaly detection, LP/MILP optimisation
BI & Visualisation	Power BI, Streamlit, dbt docs & lineage, data-quality reporting
Delivery & MLOps	FastAPI, Pydantic, Docker, CI/CD, pytest, model cards, SBOM & provenance, MCP
Languages	English (Fluent), Mandarin Chinese (Native), Japanese (JLPT N1)

SELECTED TECHNICAL PROJECTS

TfL Cycle-Hire Data Platform

2026

PySpark, dbt, DuckDB/Parquet, LightGBM, GitHub Actions, MCP

[tfl-data-engineering](#) · [Live demo](#)

- Unified 41.4M London cycle-hire journeys across 5 incompatible schema eras into one modelled warehouse (PySpark ingest to dbt to DuckDB/Parquet), with 63 dbt tests for quality and reproducibility.
- Trained a LightGBM demand forecast and quantified disruption impact ($1.42\times$ median demand, 95% CI 1.24–1.61), exposing results through a read-only MCP server for LLM/agent queries.
- Runs the full pipeline daily on GitHub Actions with no warehouse or cloud spend — a deliberate warehouse-complexity and cloud-cost decision, fully reproducible end to end.

Responsible Neobank Growth Platform

2026

Python, dbt, DuckDB/BigQuery, FastAPI, Cloud Run, scikit-learn

[responsible-neobank-growth](#) · [Live dashboard](#)

- Built a governed event-to-interface dbt/BigQuery analytics-engineering platform on synthetic neobank data, with 217 dbt data tests and 494 pytest tests plus known-truth reconciliation.
- Benchmarked full-vs-incremental warehouse builds honestly: the incremental path cut compute $\sim 63\%$ and partitioning read $\sim 524\times$ fewer bytes, holding cloud spend near £0.21 against a £10 cap.
- Developed a calibrated activation-scoring workflow (explainability, model card, FastAPI on Cloud Run) and absorbed a marketing-measurement methodology (MMM, incrementality) as documented case studies.

England & Wales Housing Decision-Support Platform

2026

dbt, Dagster, Python, FastAPI, Next.js

[england-wales-housing-decision-support](#) · [Live site](#)

- Combined 9 authoritative and open-data sources into 5 explainable suitability indicators across 7,264 England & Wales areas, orchestrated with Dagster and validated by 222 dbt data tests plus 2 dbt unit tests.
- Served the indicators through a public API and web app, with lineage and assumptions documented so each score can be inspected and challenged.
- Built as an honest portfolio/reference project (prior art exists) — the reproducible data engineering, not the product, is the point.

Community Energy Flexibility Optimiser

2026

Python, LP/MILP, Pydantic→OpenAPI→TS, Snowflake, Power BI

[community-energy-flex](#) · [Live app](#)

- Built an LP/MILP load optimiser on live UK grid data, typed end-to-end (Pydantic to OpenAPI to TypeScript) with a live API and web front end.
- Added a conditional ex-post retrospective that recomputes scheduled vs baseline windows against altered carbon curves — a synthetic stress test, framed explicitly as illustrative rather than realised-savings evidence.
- Governed data access with Snowflake row-access policies and surfaced results in Power BI for non-technical stakeholders.

EDUCATION

MSc Artificial Intelligence, University of Sheffield

Sep 2025 — Sep 2026

Core modules: Scalable Machine Learning, Natural Language Processing, Parallel Computing with GPUs, Machine Learning, Data Science, Text Processing

Sheffield, UK

BSc Computer Science, Queen's University Belfast

Sep 2021 — Jun 2024

Core modules: Data Structures and Algorithms, Software Engineering, Advanced Computer Architecture, Cloud Computing Belfast, UK